

The vertex / MPM interface: particle-to-surface contact

prototype/ecm/test_03_mesh_contact.py · log/okuda_ECM/03_mesh_contact · 10 August 2026

1. The problem, and why the obvious scheme does not fit

The epithelium is a triangulated surface with no mass; the matrix and the basement membrane are MPM continua on a background grid. Every mechanism between them — adhesion, contact, load transmission — has to cross that boundary, and in this prototype it never has: the epithelium is a recorded surface that pushes and is never pushed back.

Outside biology the problem is standard, and it is called FEM–MPM coupling. The natural scheme, *grid-node* coupling (CFEMP, Lian et al. 2011), resolves contact by comparing the two bodies’ velocities at grid nodes they share, and it requires the mesh and the grid to be *comparable in size*. Ours are not. A cell is $0.73 \Delta x$ and the basement membrane $0.1 \Delta x$, so both bodies fall inside one cell, the grid hands them a single mass-weighted velocity, and they are welded: this is exactly what runs 142–151 measured ten times over. *Particle-to-surface* coupling (ICFEMP, Chen et al. 2015) removes the restriction by testing a particle against a mesh *face* rather than against a node, and that is what is implemented here.

2. The interface, in equations

For particle p under face f with outward normal $\hat{\mathbf{n}}$, let $\mathbf{w}$ be the barycentric weights of p ’s projection on f and $z_f = \sum_i w_i z_i$ the interpolated surface. The penetration, the normal penalty, and the regularised Coulomb friction are

$$d_p = [(\mathbf{x}_p - \mathbf{x}_f) \cdot \hat{\mathbf{n}}]_+, \quad \mathbf{f}_p^n = -k d_p \hat{\mathbf{n}}, \quad (1)$$

$$\Delta \mathbf{v}_t = (\mathbf{v}_p - \sum_i w_i \mathbf{v}_i)_{\perp \hat{\mathbf{n}}}, \quad \mathbf{f}_p^t = -\min\left(\mu k d_p, \mu k d_p \frac{|\Delta \mathbf{v}_t|}{\varepsilon}\right) \frac{\Delta \mathbf{v}_t}{|\Delta \mathbf{v}_t|}, \quad (2)$$

and the reaction is distributed to the face’s vertices by the same weights, which is what makes the exchange conservative by construction:

$$\mathbf{f}_i = -w_i (\mathbf{f}_p^n + \mathbf{f}_p^t), \quad \text{so} \quad \sum_p \mathbf{f}_p + \sum_i \mathbf{f}_i = \mathbf{0}. \quad (3)$$

ε is a slip velocity, not a fudge: it is the speed below which a contact is treated as stuck, so a resting contact does not chatter.

The one constraint that is not optional. The contact force reaches the particle as a body force, integrated at Δt , so it is stable only while $\Delta t \sqrt{k/m} < 1$, i.e.

$$k < k_{\max} = \frac{m}{\Delta t^2}, \quad m = \rho \Delta x^3 / \text{ppc} = 9.5 \times 10^{-7}, \quad k_{\max} = 95. \quad (4)$$

The first version of this rig used $k = 3 \times 10^4$, three hundred times past it. The stiffness is therefore written as a *fraction* of $k_{\max}$ — here 5%, so $k = 0.238$ — which is the same discipline the flat sheet needed after it NaN’d at $k = 6 \times 10^3$.

3. As a Plexus2 container

The contact is a *relation*, so it is an edge set and not a force field on a body:

object	kind	carries
<code>contact</code>	edge set, pre: <code>mpm_particle</code> , post: <code>vertex</code>	d_p , $\mathbf{w}$, bound flag
<code>mesh_contact</code>	Lateral on <code>contact</code>	returns a delta to <i>both</i> endpoints
<code>contact_rebuild</code>	Rewire on <code>contact</code>	which particle is under which face

Two placements matter. `mesh_contact` returns `{mpm_particle: f_p, vertex: f_i}` from one call, so Eq. (3) cannot be half-implemented — which is precisely how the spheroid’s adhesion lost its reaction. And it belongs *inside* the `substep_dt` block, recomputed at the substep positions: at frame level a stiff contact is integrated at Δt_{frame} and the ceiling in Eq. (4) falls by $(\Delta t_{\text{frame}} / \Delta t_{\text{sub}})^2 = 400$.

4. The rig, and what it measures

A flat patch of 17×17 vertices with four-neighbour springs, pressed at a prescribed speed into a block of 81,202 MPM particles; mesh cell $1.68 \Delta x$, no gravity. Three variants, because one of them alone cannot separate the interface from its boundary conditions: the block free, the block trapped against a rigid floor, and the same with a hole cut in the surface. The floor is a *grid* boundary condition — the downward component of the grid velocity is zeroed below the plane, which is how MPM states an obstacle — and the hole is a real hole: 21 vertices inside $r = 0.06$ are removed, the 52 springs touching them go with them, and every quad with a missing corner stops being a contact face, so the material under it has nowhere to go but through.

run		momentum residual	penetration	slip, $\mu = 0.4$ / $\mu = 0$	compression
03	free block	1.5×10^{-7}	0.67 cells	—	—
03b	rigid floor	1.6×10^{-7}	1.47 cells	0.069 / 0.368	15.0%
03c	floor + hole	1.4×10^{-7}	1.56 cells	0.088 / 0.344	14.4%

The momentum residual is float32 machine precision in all three, so the reaction of Eq. (3) is really returned — the test a one-way coupling fails by construction, and the one the spheroid’s adhesion would fail today. **Friction changes the slip four- to fivefold**, so the contact is not MPM’s automatic weld wearing a coefficient; the free-block run cannot say this, because its slip is measured over the frames that have a contact and it lifts off before enough of them accumulate. **Penetration is bounded but is one to one and a half cells**, which is a lot: the fix is a stiffer penalty with a smaller substep, or barrier contact. And the floor is what turns an indentation into a confined compression — with nowhere to go the block loses 15% of its height, and the hole gives that back slightly (14.4%) because material escapes through it.

The particles are coloured by $|J - 1|$, the volumetric strain, on one 99th-percentile scale taken over the whole run. That scale cannot be known before the run, so the renderer steps the physics once, keeps the drawn frames, and sets the scale from them — the alternative, a value guessed in advance, is what made the falling-block movie a solid saturated colour.

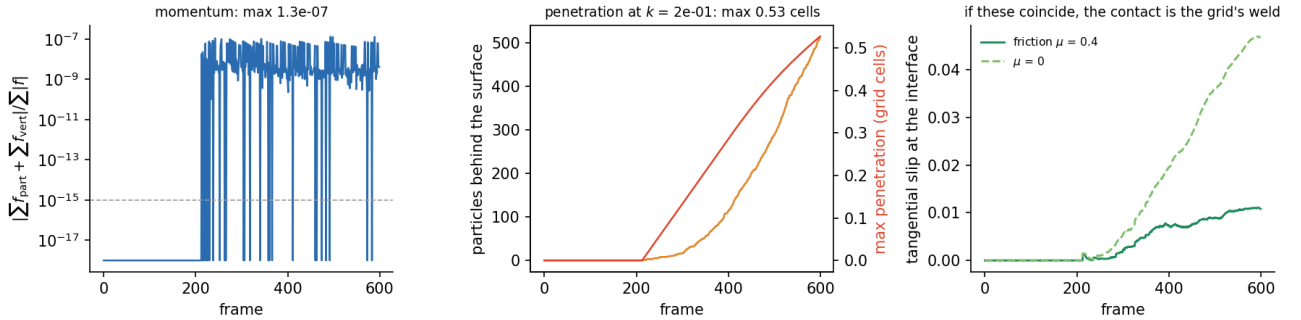


Figure 1: Run 03. Left: the momentum residual, which is the test that a one-way coupling fails by construction. Middle: how many particles are behind the surface and how deep, in grid cells. Right: tangential slip with friction and without — if these two coincided, the “contact” would be MPM’s automatic weld wearing a friction coefficient.

Two failures the rig produced before it produced a result, both worth keeping. Under gravity the block *fell out from under the patch* — its floor is twenty cells below its underside, so it was in free fall and the two bodies never met (block top $0.5800 \rightarrow 0.5728$ while the patch went $0.5850 \rightarrow 0.5778$). And a patch of independent vertices is not a mesh: pressing its rim left the interior behind, so the lattice needed edge springs before a press could load anything.

5. What this does and does not license

It licenses the interface: a surface and a continuum can exchange force conservatively at a resolution where grid-node coupling cannot, with slip that is a material property rather than a numerical accident, and it survives an obstacle behind the material and a hole in the surface without either being special-cased. It does not license the geometry — the patch is flat and axis-aligned, so face lookup is $O(1)$ and a real mesh needs a BVH — and it does not license the accuracy: half a grid cell of penetration is a lot, and the fix is either a stiffer penalty with a smaller substep or the barrier formulation (BFEMP, Li et al. 2022), which forbids penetration outright at the cost of an implicit solve. The next question is whether the same operator survives a *curved, moving* mesh, which is the spheroid.

6. The spec this argues for

Everything above, plus the archived line and the literature the note beside this one records, points at one container. It is chosen so that the *expensive* fidelity is the fidelity we cannot do without, and the cheap fidelity is taken in full.

What resolution cannot buy. A basement membrane is 100 nm on a tissue of 100 μm . To carry it as a resolved MPM body – seven cells through its thickness, which is what the pillar rig needed – costs $\Delta x \approx 3 \times 10^{-4}$ and a grid of 3500^3 : four times 10^{10} cells, for one sheet. So the sheet is *codimensional*: its thickness enters as a parameter (a rest length, Eq. ??), never as resolved geometry, which is the move thin-shell MPM and C-IPC both make. For the same reason its adhesion is an explicit relation and not grid contact: below a cell the grid welds, and runs 142–151 measured that ten times.

	representation	why not finer	cost
epithelium	vertex model, replayed	pass 1 and pass 2 cannot share a world	already paid
basement membrane	MPM particles, codim-1, thickness as ℓ_0	3500^3 grid to resolve it	45k particles
matrix	MPM fibres, $\Delta x = 1/64$	the stress front is resolved at $1/64$	140k particles
plaque	edge set $\text{cell} \rightarrow \text{bm}$, $\Sigma \approx 7T$	resolved plaques need the same 3500^3	4–20k edges
fibril	edge set $\text{bm} \rightarrow \text{ecm}$	—	4k edges

The schedule, and the one placement that matters. `plaque_pull` and `fibril_pull` go *inside* the substep block, with the MPM cycle; at frame level a stiff relation is integrated at Δt_{frame} and its stability ceiling is $400\times$ lower, which is the whole difference between run 121 and run 133. The mass balance and the proteolysis stay at frame level, where their timescale is hours:

$$\text{aggregate} \rightarrow \text{seed} \rightarrow \underbrace{\left[\text{plaque_pull}, \text{fibril_pull}, \text{mpm_strain}, \text{scatter}, \text{grid_update}, \text{gather} \right]}_{\Delta t_{\text{sub}} = 2 \times 10^{-4}} \rightarrow \text{bm_secrete}, \text{bm_degrade}, \text{e}$$

The fidelity that is nearly free, and is therefore not optional. A per-particle areal density with production and removal (Eq. ??) at the measured half-life of 3–10 hours — one scalar per particle, and the mechanism without which a sheet cannot enclose a tripling radius at all, as run 155 showed elasticity alone cannot. A strain-dependent modulus (Eq. ??), one line in the constitutive law, against a sheet that reaches $\lambda = 2.4$. A stiffness ratio of 10–100 between sheet and stroma, which costs nothing but a second parent set. And punctate adhesion at the measured spacing, which is an edge list rather than a field.

What it costs and what it still does not buy. The added operators are $O(N)$ scalars and two edge lists, so a 402-frame run stays where the current line is: roughly 45 minutes on one A6000 at 185k particles and $n_{\text{grid}} = 64$. What it does not buy is two-way mechanics with the epithelium — the tissue is still a replay, so it drives and is never driven. The operator in this note is exactly the piece that removes that limitation, and the next experiment is the one that puts a *curved, moving* mesh on the same contact: the spheroid.